A sales report that tells you where to look

Portfolio case study | Python, SQLite and Excel | Kaggle Superstore sample

The business question

When the next sales export arrives, what changed, which segments explain the change, and which source products should someone review first? This project automatically turns monthly CSVs into a written sales review, a prioritized follow-up list and an Excel workbook.

What the sample actually shows

For **December 1-30, 2018 versus December 1-30, 2017**, sales fell from **95,007.35 to 83,030.39 (-12.6%)**. Orders rose from **167 to 223 (+33.5%)**, while average order value fell from **568.91 to 372.33 (-34.6%)**. These are source monetary units; the mirror does not specify currency.

Start with these review leads

East: sales declined 7,961.58. **Office Supplies:** sales declined 7,265.53. Both recorded more orders but lower sales per order. The generated review CSV ranks five declining segments and supplies the three product IDs with the largest declines inside each segment.

For example, inspect East product **FUR-CH-10002331** against its prior-period orders. Confirm export coverage and product mix before deciding on a response. Categories and regions overlap, so their declines cannot be added together as total lost sales.

The arithmetic behind the finding

At the previous average order value, the extra orders contribute +31,858.75. The average-order change contributes -43,835.72. Together these reconcile to the sales change of -11,976.96. This is an arithmetic decomposition, not proof of a cause, stock shortage or margin loss.

Run it with ordinary Python

```
python refresh.py --summary-only
```

The standard-library path returns **sales-review.html**, **review_actions.csv**, SQL extracts and a SQLite database in a dated run folder. Read runs/latest.json to locate the newest successful output. Add new monthly inputs and rerun; the report advances automatically. The optional Excel path is documented separately.

Historical sample demonstration: 9,800 lines from 48 partitions of one Kaggle file. Kaggle is not a live monthly feed. No client outcome or time saving is claimed.

Controls, handoff & limitations

A practical example for recurring spreadsheet and data-cleaning contracts

What the workbook contains

Sales review: matched day-of-month comparisons, category/region drivers, arithmetic bridge and ranked product review leads. **Monthly report:** full observed-month totals and trend. Its December YoY is -13.3% because its 2017 baseline includes December 31; the matched review excludes that day. **Monthly detail:** period formulas and coverage. **Category review:** saved notes keyed by month/category.

Notes and refresh behavior

Export Category review as CSV UTF-8 and run **python src/import_notes.py exported-review.csv**. The importer merges annotations, keeps older keys and ignores untouched template rows. Reviewed requires both owner and note. Refreshing reads the separate notes input; it does not overwrite it. Keys absent from the current input appear in unmatched_notes.csv.

Validation and publication

Identical repeated Row IDs are skipped; conflicting IDs, missing headers, invalid dates, dates in the wrong filename month, inconsistent order dates and unsupported amounts stop the run. Every accepted row retains its source filename and row number. A missing month is unavailable; an explicit header-only monthly file is known zero. Six controls reconcile source rows and SQL totals. Each successful version contains its input hashes, metrics and workbook checks.

Evidence

Thirty-two automated tests cover ingestion, notes, publication, matched cutoffs, zero/missing baselines, leap dates, review ranking, source product IDs, arithmetic reconciliation, HTML escaping and the portable Python path. Thirteen real-engine replay checks cover a new month, retained notes, a repeat and a rejected conflict. Independent original-source and read-only XLSX checks verify the delivered outputs.

Source and operating limits

Downloaded Kaggle dataset: rohitsahoo/sales-forecasting. The data card labels the dataset GPL 2. The project records the exact source SHA-256 and supplies selected original columns as reproducible monthly inputs. The source is a Superstore sample covering Jan 3, 2015-Dec 30, 2018. Currency is unspecified by the mirror, so the report preserves source monetary units and four-decimal precision. Displayed amounts use two decimals.

Runtime and review

The useful HTML review, prioritized CSV, SQLite database and tests need only Python 3.10+. Optional Excel generation requires Node.js and the compatible artifact-tool runtime in SETUP.md. The delivered workbook can be opened independently. Formula recalculation was checked in the authoring engine; native desktop Excel was unavailable. Notes need CSV import before refresh; there is no scheduler or background watcher.

Kaggle dataset and license label | Complete source, test evidence and refresh guide accompany this briefing.